

Targeting cancer stem cells with sulforaphane, a dietary component from broccoli and broccoli sprouts.

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Many studies have supported the protective effects of broccoli and broccoli sprouts against cancer. The chemopreventive properties of sulforaphane, which is derived from the principal glucosinolate of broccoli and broccoli sprouts, have been extensively studied. Recent research into the effects of sulforaphane on cancer stem cells (CSCs) has drawn lots of interest. CSCs are suggested to be responsible for initiating and maintaining cancer, and to contribute to recurrence and drug resistance. A number of studies have indicated that sulforaphane may target CSCs in different types of cancer through modulation of NF- κ B, SHH, epithelial-mesenchymal transition and Wnt/ β -catenin pathways. Combination therapy with sulforaphane and chemotherapy in preclinical settings has shown promising results. In this article, we focus on the effects of sulforaphane on CSCs and self-renewal pathways, as well as giving a brief review of recent human studies using broccoli sprout preparations.